

# **Shelter as Infrastructure for Human Survival, Health, Dignity, and Climate Resilience**

*An Integrative Review and the SHELTER Systems Framework*

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## **Author note**

This manuscript is a publication-formatted scholarly draft. It does not report a clinical trial, field experiment, or new population dataset. The SHELTER framework and all conceptual figures are author-developed syntheses and require empirical validation before use as a regulatory or certification standard.

## Abstract

Shelter is commonly reduced to a roofed physical object, yet its effects are produced by a larger system of tenure, services, location, social relations, health conditions, environmental exposure, governance, and time. This integrative review synthesizes evidence and guidance from public health, human rights, humanitarian response, building science, disaster-risk reduction, climate adaptation, and urban policy. The review asks three questions: what functions must shelter perform; how should emergency and long-term shelter be connected; and which performance domains can support context-sensitive decision-making? Current evidence shows that shelter inadequacy operates through multiple pathways, including exposure to heat and cold, damp and indoor pollutants, crowding, injury hazards, insecurity of tenure, unaffordability, weak access to water and sanitation, social isolation, and hazard-prone location. Humanitarian guidance establishes essential minimums for safety, dignity, space, participation, and technical quality, but minimum emergency indicators should not be treated as long-term adequacy targets. Climate change increases heat, flood, storm, fire, and displacement risks, while the buildings and construction sector remains a major source of carbon emissions and material extraction. To integrate these concerns, the paper proposes the SHELTER framework: Security of tenure and protection; Health and habitability; Essential services and energy; Location, livelihoods, and social connection; Thermal, structural, and hazard resilience; Equity, accessibility, participation, and cultural fit; and Resource efficiency, repairability, and recovery. A weighted score can support comparison among locally feasible options, but non-negotiable life-safety and rights constraints must be applied as hard gates. The main conclusion is that effective shelter policy should prioritize safe occupancy, household agency, repair and upgrading where feasible, service and tenure integration, climate-responsive design, and measurable pathways from immediate protection to durable recovery.

**Keywords:** adequate housing; emergency shelter; humanitarian response; public health; climate resilience; disaster recovery; tenure security; building sustainability; displacement

## Highlights

- Shelter is a coupled physical, social, legal, service, and environmental system rather than a standalone product.

- Emergency minimums are life-saving floors and should not become permanent ceilings for displaced or low-income populations.
- Health, climate, tenure, accessibility, and livelihood performance should be evaluated together.
- The author-developed SHELTER framework provides seven domains and a gated scoring method for comparing context-specific options.

## 1. Introduction

Shelter is one of the oldest technologies of survival and one of the most consequential institutions of society. It mediates exposure to weather, pathogens, violence, pollution, noise, injury, and social stress; it anchors access to work, education, care, infrastructure, identity, and political recognition. The Universal Declaration of Human Rights includes housing within an adequate standard of living, while subsequent human-rights interpretation emphasizes that adequacy extends beyond possession of a structure (United Nations, 1948; Committee on Economic, Social and Cultural Rights, 1991). In ordinary policy, the issue is often named housing; in humanitarian practice, shelter commonly denotes immediate and transitional forms of accommodation after disaster, conflict, eviction, or displacement. These fields overlap but are too often planned separately.

The scale of the challenge is substantial. UN-Habitat's World Cities Report 2026 estimates that as many as 3.4 billion people lack secure, safe, and adequate housing, including more than one billion people in informal settlements and slums (UN-Habitat, 2026). At the end of 2025, one in every 70 people worldwide was forcibly displaced; the global total included 41.6 million refugees and 68.7 million people internally displaced by conflict or violence (UNHCR, 2026a). These categories overlap with poverty, climate exposure, land insecurity, and inadequate services. Meanwhile, the production and operation of buildings are deeply implicated in environmental change: UNEP reports that buildings and construction account for about 37% of global carbon dioxide emissions and nearly half of material extraction (UNEP, 2026).

These statistics reveal a central paradox. Shelter is indispensable for adaptation and public health, yet poorly located, poorly constructed, unaffordable, or carbon-intensive shelter can increase vulnerability. Airtight envelopes can reduce energy demand but worsen indoor air if ventilation is neglected. Concrete may provide durability and fire resistance but can carry high embodied emissions. Rapid imported units may provide immediate cover but weaken local livelihoods, fit poorly with climate or culture, and

become long-lived without adequate services. Dense settlements can improve access and reduce transport demand while also increasing fire spread, heat, crowding, and infectious-disease risks when infrastructure is deficient. Therefore, shelter cannot be optimized through a single variable such as unit cost, floor area, assembly speed, or energy efficiency.

This paper develops an interdisciplinary account of shelter as human infrastructure. It first clarifies definitions and methods, then reviews rights, health, emergency response, climate and disaster resilience, sustainability, economics, and governance. It concludes with an original conceptual contribution: the SHELTER framework, which organizes seven domains for evaluating shelter pathways across emergency and long-term settings.



**Core idea:** shelter performance emerges from interactions across all layers.

*Author-developed systems synthesis. A sound roof alone is insufficient without services, settlement support, and enabling governance.*

**Figure 1. Shelter as a nested human-protection system. The diagram is an author-developed synthesis: household outcomes are produced through interactions among the dwelling, essential services, settlement context, and governance.**

## **2. Review Scope and Method**

### **2.1 Review design**

The study is an integrative narrative review and conceptual synthesis rather than a systematic review or meta-analysis. The approach was selected because “shelter” spans heterogeneous bodies of evidence: legal standards, public-health guidelines, humanitarian minimum standards, engineering and building science, urban policy, disaster-risk reduction, climate assessment, and social-science research. The objective was to identify convergent performance principles, tensions among disciplines, and gaps that prevent shelter decisions from being evaluated as a whole.

### **2.2 Search and source selection**

Searches were conducted through July 2026 using combinations of the terms shelter, adequate housing, housing and health, emergency shelter, displacement, humanitarian standards, thermal comfort, climate resilience, disaster recovery, tenure security, accessibility, building emissions, and informal settlements. Priority was given to authoritative international guidance from WHO, UN-Habitat, UNHCR, the Sphere Association, IPCC, UNEP, UNDRR, and United Nations human-rights bodies. Peer-reviewed studies were selected when they contributed a conceptual model, synthesized health evidence, evaluated humanitarian shelter conditions, or examined the contextual limits of standards. Sources were included if they addressed one or more of four levels: household and user experience; dwelling performance; settlement and service systems; or policy, rights, and environmental impacts.

### **2.3 Analytical procedure**

Evidence was coded into recurring functional domains: protection, health, services, spatial access, hazard performance, inclusion, tenure, affordability, environmental burden, participation, and recovery. The domains were compared across permanent housing and humanitarian shelter guidance. Redundant concepts were merged, while concepts that create non-substitutable obligations, such as structural life safety and protection from forced eviction, were retained as constraints. The final seven-domain SHELTER framework was created through abductive synthesis: it is grounded in the reviewed literature but is not itself an established standard.

## 2.4 Limitations

The review did not search every bibliographic database, apply a formal risk-of-bias instrument, or quantify pooled effect sizes. International indicators are affected by differences in definitions, reporting systems, and reference years. Evidence remains geographically uneven, particularly for informal, mobile, indigenous, and protracted-displacement settings. The proposed framework should therefore be treated as a transparent hypothesis and decision aid requiring local adaptation and empirical validation.

## 3. What Counts as Shelter?

A useful definition must be broad enough to connect emergency and permanent contexts without erasing their differences. In this paper, shelter is defined as a place-based arrangement that enables people to occupy space with protection, dignity, essential services, and a practicable degree of stability. The arrangement may include a private dwelling, hosted accommodation, rented rooms, collective centres, temporary structures, repaired buildings, incremental housing, or permanent reconstruction. Shelter is therefore both material and institutional: it includes the envelope and site, the rules that permit occupancy, the services that make occupancy healthy, and the social relations that make the place usable.

***Table 1. Shelter contexts, primary purposes, and characteristic risks.***

| <b>Context</b>                | <b>Primary purpose</b>                                                   | <b>Typical duration</b> | <b>Characteristic risks if poorly managed</b>                                                            |
|-------------------------------|--------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------|
| Routine or permanent housing  | Long-term residence, family life, asset formation and access to the city | Years to generations    | Unaffordability, unsafe construction, energy poverty, eviction, segregation, unhealthy indoor conditions |
| Homelessness or night shelter | Immediate refuge from street exposure and violence                       | Hours to months         | Congregate infection, lack of privacy, institutional exclusion, unsafe discharge back to homelessness    |
| Evacuation centre             | Short-term safety during an anticipated or active hazard                 | Hours to weeks          | Overcrowding, inaccessible facilities, interruption of schooling, inadequate privacy and safeguarding    |

| Context                                      | Primary purpose                                                       | Typical duration     | Characteristic risks if poorly managed                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------|
| Emergency humanitarian shelter               | Rapid protection after disaster, conflict or displacement             | Days to months       | Exposure, fire, heat/cold stress, poor siting, weak services, aid dependency                             |
| Transitional or temporary shelter            | Intermediate stability while recovery pathways are resolved           | Months to years      | Temporary structures becoming permanent, unresolved tenure, high lifecycle cost, material waste          |
| Repair, retrofit or incremental/core housing | Restore or progressively improve an existing or future permanent home | Months to decades    | Unsafe repair, exclusion of renters, inadequate technical support, lock-in of hazards                    |
| Reconstruction or planned resettlement       | Durable recovery and risk reduction                                   | Years to generations | Loss of livelihoods and social networks, debt, land conflict, maladaptation, carbon-intensive rebuilding |

### 3.1 Adequacy is multidimensional

The human-rights concept of adequate housing includes legal security of tenure, availability of services and infrastructure, affordability, habitability, accessibility, location, and cultural adequacy (Committee on Economic, Social and Cultural Rights, 1991). This formulation has two important implications. First, a technically robust unit can still be inadequate if it is unaffordable, inaccessible, isolated from livelihoods, or occupied under threat of eviction. Second, adequacy is not identical to uniformity: culture, climate, household composition, disability, livelihood practices, and tenure systems change what a workable shelter must provide.

### 3.2 Minimum, adequate, and aspirational performance

Standards serve different functions. A minimum standard identifies a floor below which urgent harm is likely or rights are clearly compromised. An adequacy standard describes conditions compatible with health, dignity, and sustained daily life. An aspirational standard drives improvement toward low-carbon, regenerative, or exceptionally inclusive performance. Confusing these categories can create two errors: treating emergency minimums as acceptable permanent living conditions, or applying

idealized requirements so rigidly that urgent assistance is delayed. Good practice uses minimums to protect life immediately while planning measurable progression toward adequate and durable conditions.

#### **4. Shelter, Rights, Tenure, and Protection**

Housing appears in Article 25 of the Universal Declaration of Human Rights as part of a standard of living adequate for health and well-being (United Nations, 1948). General Comment No. 4 of the Committee on Economic, Social and Cultural Rights established the influential interpretation that the right is not merely a right to any roof, but a right to live somewhere in security, peace, and dignity (Committee on Economic, Social and Cultural Rights, 1991). This rights-based understanding is especially important where shelter interventions affect land claims, renters, informal occupants, women's inheritance, indigenous territories, or people who have lost documents during crisis.

Tenure security is not synonymous with full ownership. It refers to a degree of protection against forced eviction, harassment, and arbitrary displacement. Sphere guidance similarly treats security of tenure as foundational and urges due diligence across formal, customary, rental, communal, and informal arrangements (Sphere Association, 2018). Programmes that assist only documented owner-occupiers may exclude renters, subtenants, occupants of collective land, and households headed by women. Conversely, building on disputed land without a "secure enough" arrangement can expose residents to later eviction or conflict.

Protection also includes privacy, lighting, safe access routes, fire egress, safeguarding, gender-sensitive facilities, and the ability to control entry. A shelter may be structurally sound yet unsafe for a woman walking to a distant toilet at night, a child in an unpartitioned collective centre, an older person unable to climb steps, or a stigmatized group exposed to harassment. Participation is therefore not only a procedural ideal; it is a method for discovering risks that standardized technical checklists can miss.



## **5. Shelter as a Public-Health Intervention**

Housing is a major social determinant of health. Classical reviews link poor housing with respiratory infection, asthma, injury, toxic exposure, and mental ill-health (Krieger & Higgins, 2002). WHO's housing and health guidelines synthesize evidence on crowding, low and high indoor temperatures, injury hazards, and accessibility, while also connecting housing policy to air quality, water, noise, lead, asbestos, radon, and tobacco smoke (WHO, 2018). More recent frameworks emphasize that health is shaped not only by physical defects but also by affordability, stability, neighbourhood conditions, control, and the experience of home (Swope & Hernández, 2019; Rolfe et al., 2020; Bentley et al., 2025).

### **5.1 Thermal conditions**

Shelter protects health by moderating temperature and humidity. In cold climates, inadequate insulation, heating, moisture control, and ventilation contribute to respiratory and cardiovascular stress. In hot climates, solar gain, poorly ventilated roofs, limited shade, and power insecurity can create dangerous indoor overheating. Climate change intensifies these problems by raising the frequency and severity of heat events and by increasing the likelihood that households must choose between energy costs and safe temperatures (IPCC, 2022; Li et al., 2025). Passive measures such as orientation, shading, reflective or insulated roofs, cross-ventilation, thermal mass appropriate to diurnal conditions, vegetation, and safe night ventilation can reduce risk, but their performance is climate- and context-specific.

### **5.2 Indoor air, damp, and combustion**

Indoor air quality depends on outdoor pollution, ventilation, occupancy density, damp, mould, material emissions, cooking fuels, smoking, and maintenance. Temporary shelters frequently have low volume, high occupancy, thin envelopes, and constrained cooking arrangements. A review of displaced-population shelters found that indoor air quality, thermal comfort, and overcrowding were among the most frequently identified health concerns (Conzatti et al., 2022). Interventions must avoid simplistic trade-offs: sealing air leakage without controlled ventilation can trap pollutants, while opening shelters without insect control or security can create other harms.

### 5.3 Crowding, privacy, and mental well-being

Crowding changes infectious-disease transmission, sleep, caregiving, conflict, privacy, and psychological recovery. Sphere’s emergency indicator of at least 3.5 m<sup>2</sup> of living space per person, excluding cooking, bathing, and sanitation, is an operational minimum for many emergency settings; larger allowances are identified for cold climates and some urban arrangements (Sphere Association, 2018). The handbook itself cautions that minimum space must be contextualized and reconsidered when stays extend. A numerical area cannot fully represent adequacy because layout, ceiling height, storage, household composition, partitions, external shaded space, and access to communal facilities shape whether the area supports daily life.

### 5.4 Injury, fire, and accessibility

Structural failure, falls, burns, electrocution, unsafe stairs, unguarded openings, sharp elements, and poor egress are preventable shelter hazards. Fire risk becomes acute in dense settlements that combine combustible materials, cooking flames, overloaded electrical systems, and narrow access routes. Accessibility must be treated as a core performance requirement rather than a later adaptation. Step-free routes, reachable controls, adequate turning space, visual and tactile cues, stable surfaces, accessible sanitation, and proximity to services benefit people with disabilities, older residents, injured persons, caregivers, and children.

***Table 2. Major shelter-health pathways and corresponding design or policy responses.***

| <b>Pathway</b> | <b>Examples of exposure or mechanism</b>              | <b>Potential outcomes</b>                                            | <b>Integrated responses</b>                                                                                 |
|----------------|-------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Thermal        | Cold interiors, overheating, humidity, energy poverty | Cardiorespiratory stress, heat illness, poor sleep, excess mortality | Climate-responsive envelope, shade, ventilation, safe heating/cooling, energy support, heat-health planning |

| <b>Pathway</b>           | <b>Examples of exposure or mechanism</b>                                          | <b>Potential outcomes</b>                                            | <b>Integrated responses</b>                                                                        |
|--------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Air and moisture         | Smoke, mould, damp, radon, material emissions, inadequate ventilation             | Respiratory disease, irritation, poisoning, infection risk           | Source control, dry construction, ventilation, clean cooking, maintenance, material selection      |
| Crowding and space       | High occupancy, absent partitions, inadequate storage and external activity space | Transmission, stress, conflict, impaired caregiving and learning     | Adequate area and volume, partitions, household-sensitive layouts, safe communal space             |
| Injury and fire          | Weak structures, unsafe wiring, flames, blocked egress, falls                     | Trauma, burns, disability, death                                     | Code-appropriate structure, multiple exits, fire breaks, safe energy, inspection and user training |
| Affordability and tenure | Rent burden, eviction risk, unstable occupancy, debt                              | Stress, displacement, treatment interruption, food-energy trade-offs | Rental protection, social assistance, secure-enough agreements, legal aid, affordable supply       |
| Location and services    | Floodplains, polluted sites, distant work/care, unsafe routes, inadequate WASH    | Injury, infection, lost income, isolation, gendered violence         | Risk-informed siting, infrastructure, transport, proximity planning, lighting and safeguarding     |
| Control and identity     | Institutional rules, cultural mismatch, lack of privacy or personalization        | Loss of dignity, anxiety, low uptake, social fragmentation           | Co-design, household choice, culturally appropriate space, complaint and feedback systems          |

## 6. Emergency Shelter and Forced Displacement

Humanitarian shelter must act under severe uncertainty: damaged infrastructure, disrupted markets, contested land, insecurity, seasonal hazards, limited transport, incomplete population data, and urgent exposure. UNHCR recommends rapid, cross-sectoral shelter and settlement assessment early in an emergency, linked with protection, housing-land-property issues, water and sanitation, health, and local capacity (UNHCR, 2026b). Sphere frames shelter within the rights to dignity and

assistance and requires coordination, participation, technical quality, and contextualization (Sphere Association, 2018).

### **6.1 Assistance options, not product selection**

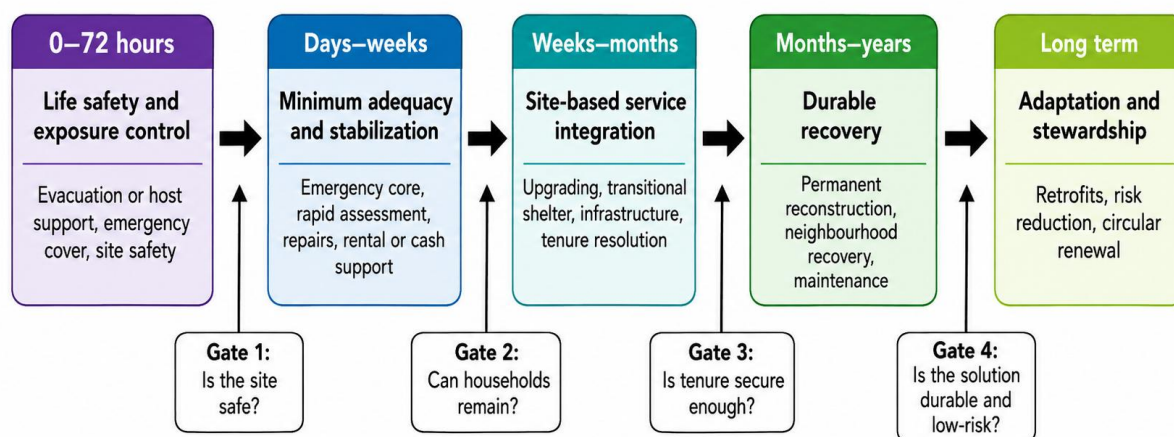
A shelter response should begin with household and settlement outcomes rather than a preferred product. Options include emergency materials, technical advice, repair grants, rental assistance, support to host families, collective-centre upgrading, cash assistance, transitional structures, core housing, and reconstruction. The optimal option depends on whether buildings can be safely repaired, markets function, land is available, households wish to return, host communities can absorb demand, and services can be restored. Imported prefabricated units can be appropriate in some conditions, but speed at the factory does not guarantee speed or suitability at the site.

### **6.2 Contextualizing minimum standards**

Sphere standards are among the most widely used humanitarian references, but research in Nairobi informal settlements shows why context matters. Patel and Chadhuri (2019) found that some non-crisis urban residents lived below selected Sphere indicators, raising difficult questions about aspirational minimums, chronic deprivation, and equity between assisted and host populations. This does not justify lowering emergency protection. Rather, it shows that shelter programmes must analyse the surrounding city, avoid creating isolated enclaves, support shared services where appropriate, and advocate for improvements that benefit displaced and non-displaced communities.

### **6.3 Protracted displacement and the temporary-permanent trap**

Temporary settlements often persist for years. Thin envelopes, minimal space, weak tenure, and deferred infrastructure then produce cumulative health and protection costs. Conversely, premature investment in permanent structures on unsafe or disputed sites can lock communities into risk. The response therefore needs an explicit time strategy: what can be upgraded, reused, relocated, expanded, recycled, or incorporated into permanent housing; who will maintain services; and what event triggers transition to a different pathway?



Phases overlap; repair, hosting and rental support may outperform imported units when local markets, tenure and capacity permit.

Author-developed conceptual response pathway for disaster, displacement and housing recovery contexts.

**Figure 2. Shelter continuum and decision gates.** The diagram is an author-developed synthesis showing how immediate exposure control should connect to stability, durable recovery, maintenance, and adaptation.

**Table 3. Comparative characteristics of common shelter assistance pathways.**

| Pathway                              | Potential strengths                                          | Principal limitations or risks                                        | Best-fit conditions                                                            |
|--------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Emergency materials or household kit | Very rapid; flexible; supports self-recovery                 | May be inadequate without skills, safe site, tools or follow-up       | Households have access to land or repairable structures and can build safely   |
| Host-family or community support     | Preserves social networks; uses existing stock; avoids camps | Overcrowding, hidden costs, safeguarding concerns, host fatigue       | Strong social ties and adequate host capacity with monitored support           |
| Rental or cash assistance            | Uses urban stock and markets; preserves choice and mobility  | Rent inflation, eviction, poor-quality units, finite subsidy          | Functioning market, tenant protection, inspection and exit strategy            |
| Collective-centre adaptation         | Fast use of existing buildings; service concentration        | Privacy, crowding, accessibility, fire and institutionalization risks | Very short-term need with safe structure, management and rapid transition plan |

| <b>Pathway</b>                 | <b>Potential strengths</b>                                      | <b>Principal limitations or risks</b>                                    | <b>Best-fit conditions</b>                                                     |
|--------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Repair and retrofit            | Often fastest durable option; preserves place and materials     | Unsafe structures, owner bias, uncertain tenure, uneven workmanship      | Damage is repairable, occupants can remain/return, technical support available |
| Transitional or modular unit   | Predictable unit; can bridge time; possible reuse or relocation | High logistics cost, cultural/thermal mismatch, temporary-permanent trap | Clear land, defined duration, service plan, reuse or conversion strategy       |
| Core or incremental housing    | Creates a durable nucleus and household agency                  | Needs land, finance, technical oversight and long-term services          | Secure-enough tenure and realistic pathway for safe expansion                  |
| Reconstruction or resettlement | Can reduce risk and deliver durable services                    | Slow, expensive, carbon intensive; may disrupt livelihoods and community | Original site is irreparable/unsafe or comprehensive recovery is feasible      |

## 7. Climate Change, Disaster Risk, and Resilient Shelter

Climate risk is produced by the interaction of hazard, exposure, and vulnerability. Homes and settlements are increasingly exposed to extreme heat, heavy precipitation, river and coastal flooding, wildfire, drought, storms, and cascading service failures. IPCC assessments emphasize that climate risks to cities, settlements, and infrastructure rise with warming and are particularly severe where vulnerability and exposure are already high (IPCC, 2022). The Sendai Framework calls for understanding risk, strengthening governance, investing in resilience, and building back better in recovery (UNDRR, 2015).

### 7.1 Risk-informed location

Siting decisions can dominate the performance of a shelter. A strong building on a frequently inundated site, unstable slope, wildfire corridor, contaminated land, or location without evacuation access remains unsafe. Yet relocation can create new harm by severing livelihoods, schooling, care networks, land relationships, and cultural

identity. Risk-informed planning must therefore compare in-situ protection, managed retreat, vertical or flood-resilient adaptation, service redundancy, and relocation through transparent participation and compensation.

## **7.2 Multi-hazard design and graceful failure**

Shelter standards should reflect locally credible hazard combinations rather than a single design event. Wind-resistant roofs need connections that do not worsen seismic behaviour; flood-resilient elevations require accessible routes and safe foundations; wildfire-resistant envelopes need defensible space and evacuation planning; highly insulated cold-climate buildings require moisture and ventilation control. “Graceful failure” is a useful principle: when design thresholds are exceeded, components should fail without disproportionate collapse, occupants should have warning and egress, and critical functions should be recoverable.

## **7.3 Passive survivability and service continuity**

Passive survivability describes the ability of a building to maintain life-supporting conditions during loss of electricity, fuel, water, cooling, or transport. It depends on envelope performance, natural ventilation where safe, shade, thermal storage, daylight, water reserves, sanitation contingencies, safe cooking, and community support. Resilience also includes maintenance capacity, local spare parts, accessible repair knowledge, and governance arrangements for shared infrastructure.

## **8. Environmental Sustainability and Circular Recovery**

Shelter is simultaneously a consumer of energy and materials and a platform for reducing future vulnerability. UNEP’s 2025-2026 global status report identifies the buildings and construction sector as responsible for roughly 37% of global carbon dioxide emissions and nearly 50% of material extraction (UNEP, 2026). Rapid urbanization and post-disaster rebuilding can therefore either lock in high-emission, resource-intensive patterns or accelerate low-carbon, resilient development.



## **8.1 Operational and embodied impacts**

Operational energy includes heating, cooling, ventilation, lighting, appliances, and water systems. Embodied impacts arise from extraction, manufacturing, transport, construction, maintenance, replacement, and end-of-life processes. The balance varies by climate, material, lifespan, and energy system. A low-embodied-carbon shelter that deteriorates quickly or requires repeated replacement may perform poorly over its lifecycle; a durable high-mass structure may be justified in some hazards but excessive in others. Lifecycle assessment should be combined with durability, adaptability, and social value rather than treated as a single carbon number.

## **8.2 Repair first, design for disassembly**

Where structures are safe enough to retain, repair and retrofit can reduce waste, preserve neighbourhoods, shorten displacement, and avoid new land consumption. New components should favour standard dimensions, reversible connections, separable material layers, non-toxic finishes, replaceable high-wear elements, and documented maintenance. Humanitarian products should have an exit pathway: reuse as a room, veranda, roof, shop, school component, storage element, or recoverable material, rather than becoming unmanaged waste.

## **8.3 Local materials and local capacity**

Local materials can reduce transport and support livelihoods, but “local” does not automatically mean safe or sustainable. Unsustainably harvested timber, unreinforced heavy masonry in seismic zones, toxic treatments, or flood-sensitive earth construction can create severe risks. Appropriate localization combines material testing, hazard-aware detailing, skills development, quality assurance, and respect for vernacular knowledge. External engineering and supply chains should strengthen local capacity rather than displace it.

## **9. Affordability, Markets, Governance, and Delivery**

Shelter affordability is not simply the purchase price of a unit. Households bear rent or debt, land costs, transport, energy, water, maintenance, transaction fees, and the opportunity cost of location. A low-rent dwelling far from work may be unaffordable



in total. Likewise, an efficient building with expensive proprietary parts can impose future maintenance burdens. Total cost of occupancy and lifecycle public cost should guide policy.

## **9.1 Supply, demand, and distribution**

Shelter crises reflect shortages, unequal access, speculative land dynamics, weak rental protection, infrastructure constraints, low incomes, conflict, disaster, and discriminatory institutions. Supply expansion is necessary in many markets but cannot alone ensure adequacy for households with the least purchasing power. Public, cooperative, community-led, rental, incremental, and private models may all contribute. The policy mix should address serviced land, planning, finance, construction capacity, vacancy, informal upgrading, homelessness prevention, and social protection.

## **9.2 Cash assistance and market safeguards**

Cash and rental assistance can increase choice, speed, and local economic activity. Their effectiveness depends on available stock, accessible information, anti-discrimination measures, quality inspection, legal agreements, dispute resolution, monitoring of rent inflation, and support for households who cannot navigate markets. Market-based delivery is a means, not an outcome; success must still be measured in safety, stability, health, dignity, and access.

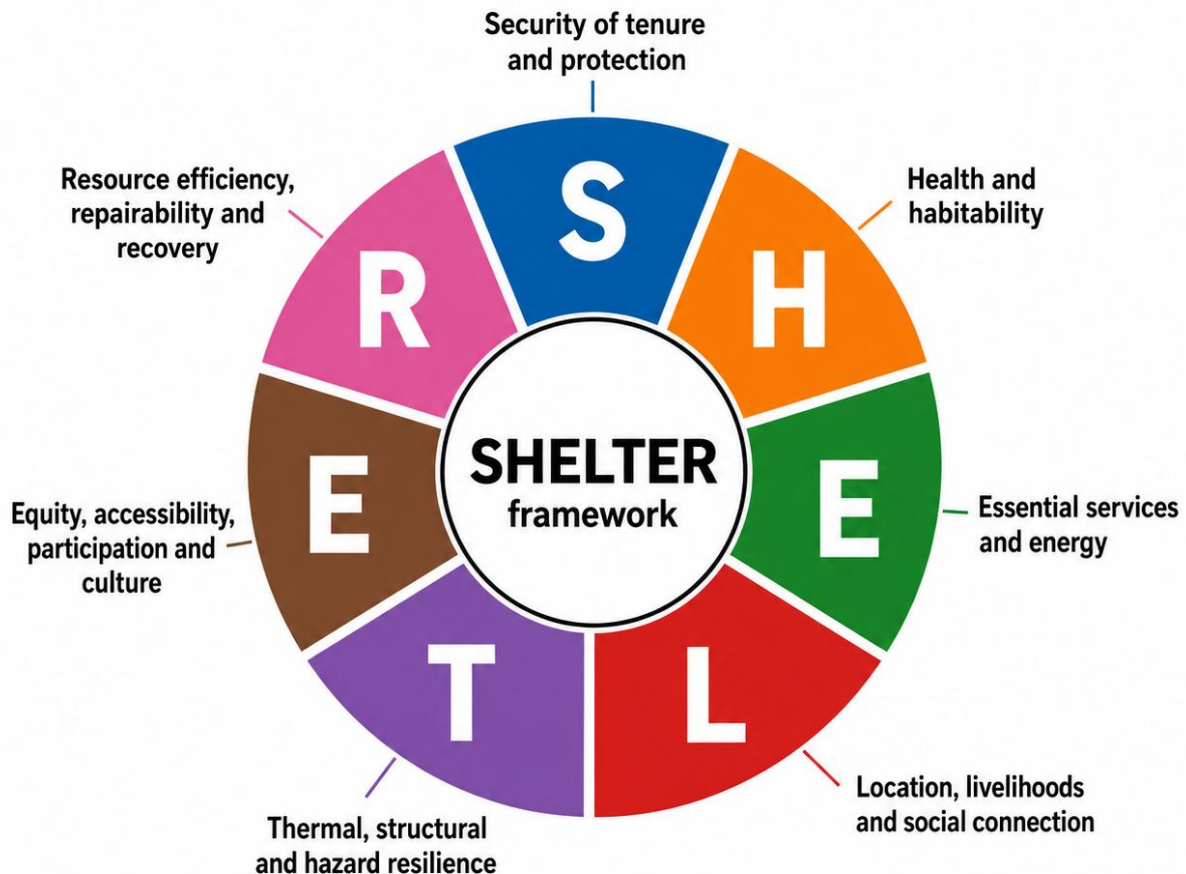
## **9.3 Accountability and co-production**

Residents are experts in household routines, cultural requirements, care practices, informal economies, and hidden protection risks. Co-production should include accessible information, meaningful options, representation of marginalized groups, transparent eligibility criteria, complaint mechanisms, feedback after occupancy, and the power to influence corrections. Participation without decision power can become extractive consultation; technical assistance without accountability can transfer risk to households.

## 10. The SHELTER Systems Framework

The review suggests that shelter options should be evaluated through seven linked domains. The acronym SHELTER is intended as a practical memory structure rather than a claim of universal completeness. Each domain can be represented by locally selected indicators. The framework is applicable to a household, building, settlement, programme, or policy, provided the unit of analysis is stated.

Use: contextual appraisal of shelter options across social, physical and service dimensions.



*Author-developed conceptual framework; intended for contextual scoring and testing, not as a universal certified standard.*

***Figure 3. The SHELTER framework. The seven domains are an author-developed conceptual synthesis and should be validated and weighted with affected communities and technical authorities.***

### 10.1 Domain definitions

- S - Security of tenure and protection: protection from forced eviction and violence; privacy; safe access; documentation; dispute resolution; control of entry.

- H - Health and habitability: indoor air, moisture, temperature, space, lighting, noise, sanitation-related conditions, injury prevention, and psychological well-being.
- E - Essential services and energy: reliable and affordable water, sanitation, drainage, waste management, clean energy, communications, and access to care.
- L - Location, livelihoods, and social connection: exposure of the site, access to work, education, transport and markets, continuity of community networks, and avoidance of segregation.
- T - Thermal, structural, and hazard resilience: code-appropriate stability, climate-responsive design, fire safety, egress, multi-hazard performance, passive survivability, and repair after shock.
- E - Equity, accessibility, participation, and cultural fit: universal access, gender and age sensitivity, household diversity, cultural adequacy, choice, participation, and fair distribution.
- R - Resource efficiency, repairability, and recovery: lifecycle emissions, responsible materials, adaptability, local skills, maintenance, disassembly, reuse, and a credible path to durable recovery.

## 10.2 Scoring with non-compensable gates

For comparative planning, each domain may receive a score  $s_i$  on a defined scale, such as 0 (unacceptable) to 5 (strong performance), and a context-specific weight  $w_i$ . A transparent aggregate score can be written as:

$$P = \sum_{i=1}^7 w_i s_i, \quad \text{where } \sum_{i=1}^7 w_i = 1. \quad (1)$$

However, simple aggregation can conceal unacceptable trade-offs. A very low-cost, low-carbon unit should not receive a favourable result if it is structurally unsafe or exposes residents to eviction. The framework therefore applies feasibility gates  $g_j$  for non-negotiable constraints, with  $g_j = 1$  when a condition is satisfied and 0 when it is not. A gated score is:

$$P^* = (\prod_{j=1}^m g_j) P. \quad (2)$$

Possible gates include immediate structural life safety, absence of an unmitigated extreme site hazard, minimum safe water and sanitation access, protection from foreseeable forced eviction during the assistance period, accessible evacuation for intended occupants, and compliance with mandatory law. A failed gate does not always mean abandoning an option; it may identify the mitigation required before implementation. Weights and thresholds should be disclosed, stress-tested, and agreed with affected groups rather than selected only by implementers.

**Table 4. Example SHELTER indicators for programme design and post-occupancy evaluation.**

| <b>Domain</b>                 | <b>Illustrative indicators</b>                                                                                                                  | <b>Evidence sources or methods</b>                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Security                      | Occupancy agreement; perceived safety; eviction incidence; privacy; safe access and lighting; functioning complaint/referral routes             | Tenure review, household survey, protection audit, incident and referral records                   |
| Health                        | Indoor temperature and humidity; ventilation/CO <sub>2</sub> proxy; damp/mould; crowding; injury hazards; sleep and self-reported health        | Sensors, inspection, household survey, clinic or surveillance data where ethical                   |
| Essential services            | Water quantity/quality and continuity; sanitation accessibility; drainage; energy affordability; clean cooking; waste collection                | Service logs, utility expenditure, WASH testing, observation, user diaries                         |
| Location                      | Travel time/cost to work, school, care and markets; hazard exposure; transport and evacuation access; social-network continuity                 | GIS, participatory mapping, mobility survey, hazard maps and local knowledge                       |
| Thermal/structural resilience | Code or technical compliance; roof and connection performance; fire separation/egress; passive survivability; time and cost to repair           | Engineering assessment, drills, scenario analysis, damage and repair records                       |
| Equity                        | Step-free access; accessible sanitation; gender/age safety; participation; choice; satisfaction by subgroup; cultural fit                       | Accessibility audit, disaggregated survey, focus groups, decision and grievance records            |
| Resource/recovery             | Embodied and operational carbon; local/reused content; maintenance burden; adaptability; disassembly/reuse rate; recovery milestone achievement | Lifecycle assessment, bills of materials, maintenance logs, material-flow and programme monitoring |

### 10.3 Decision sequence

A practical application can follow six steps: define the affected population and time horizon; identify non-negotiable legal and life-safety gates; generate multiple

assistance pathways rather than a single product; score the pathways across the seven domains using disaggregated evidence; test sensitivity to weights, hazard scenarios, duration, and market change; and establish post-occupancy monitoring with authority and budget for correction. The framework is most valuable when it exposes a trade-off early, such as a fast solution with poor thermal performance, a resilient site with weak livelihood access, or an affordable rental programme with insecure tenure.

## 11. Research and Implementation Agenda

Shelter research remains fragmented by discipline and by the divide between chronic housing deprivation and humanitarian emergency. A more cumulative evidence base requires shared outcomes, longitudinal follow-up, and ethical data systems that do not increase surveillance or tenure risk. The following priorities emerge from the review.

***Table 5. Priority research questions and suitable study designs.***

| Priority                       | Research question                                                                                                           | Suitable approaches                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Minimum-to-adequate transition | How quickly do emergency minimums become harmful when stays are prolonged, and which upgrades deliver the greatest benefit? | Longitudinal cohorts, stepped-wedge upgrades, post-occupancy evaluation, participatory indicators       |
| Heat and compound hazards      | Which low-cost passive measures maintain safe indoor conditions during heat plus power loss, smoke, flood or insecurity?    | Field monitoring, full-scale tests, calibrated simulation, extreme-event natural experiments            |
| Tenure and health              | How do secure-enough agreements, rental protection and legal aid affect health, investment and displacement?                | Quasi-experiments, administrative linkage with safeguards, realist evaluation                           |
| Accessibility in crisis        | Which designs and delivery models work for people with diverse mobility, sensory, cognitive and care needs?                 | Co-design, usability testing, inclusive drills, disaggregated outcome analysis                          |
| Lifecycle and circularity      | When do repair, local materials, modularity, reuse or reconstruction minimize total emissions and social cost?              | Consequential lifecycle assessment, material-flow analysis, lifecycle costing, comparative case studies |

| Priority                     | Research question                                                                                                 | Suitable approaches                                                                            |
|------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Local markets and equity     | When does cash or rental support improve choice without inflation, exclusion or quality decline?                  | Market monitoring, randomized or quasi-experimental programme evaluation, tenant surveys       |
| Governance and participation | Which forms of resident power lead to safer design corrections, stronger maintenance and more durable recovery?   | Process tracing, comparative governance studies, participatory action research                 |
| Framework validation         | Does the SHELTER framework predict health, satisfaction, stability, recovery speed and environmental performance? | Delphi refinement, multi-country validation, reliability testing, prospective programme trials |

## 12. Discussion

The reviewed evidence converges on a systems interpretation of shelter. Physical quality matters greatly, but the dwelling is only one node in a chain that includes land, services, mobility, livelihoods, governance, household control, and environmental change. This explains why technically identical units can produce different outcomes across sites and households. It also explains why shelter policies that optimize unit delivery can underperform when they neglect maintenance, tenure, public space, safety, or access to the city.

The integration of emergency shelter and housing policy is especially important. Humanitarian actors possess operational expertise in rapid assessment, protection, minimum standards, and crisis logistics. Housing and urban-policy institutions possess longer-term authority over land, infrastructure, finance, codes, and neighbourhood development. Protracted displacement exposes the cost of leaving these systems disconnected. The shelter continuum in Figure 2 should therefore be institutional as well as physical: emergency decisions should preserve future choices, and development systems should be prepared to absorb and improve crisis responses.

The SHELTER framework also clarifies that some trade-offs are legitimate while others are not. Costs, speed, floor area, embodied carbon, and durability can be balanced within context. Structural collapse risk, foreseeable forced eviction, inaccessible evacuation, or exposure to severe violence cannot simply be averaged against strengths elsewhere. The gated scoring approach formalizes this distinction. Its

weakness is that gates and weights can still be manipulated; public disclosure, resident participation, independent review, and post-occupancy data are therefore essential.

A final implication concerns innovation. Shelter innovation is often presented as a new object: a folding unit, novel material, digital platform, or prefabricated system. Product innovation can be useful, but system innovations may be more consequential: pre-disaster tenure records, rapid building assessment, repair financing, inclusive rental registries, mutual-aid hosting, adaptable codes, local component standards, heat-health protocols, and procurement that values lifecycle outcomes. Innovation should be judged by measurable improvement in the seven SHELTER domains, not novelty alone.

### **13. Conclusion**

Shelter is the infrastructure through which people convert space into safety, health, privacy, belonging, livelihood, and recovery. A roof and walls are essential, but they are insufficient without secure occupancy, essential services, safe location, healthy indoor conditions, accessibility, cultural fit, and the capacity to withstand and recover from hazards. The global housing crisis, forced displacement, and climate change make fragmented approaches increasingly untenable.

This review proposes five practical conclusions. First, assess shelter as a nested household-dwelling-settlement-governance system. Second, treat emergency minimums as temporary life-saving floors and define an upgrade pathway from the beginning. Third, consider repair, hosting, rental support, and incremental approaches before defaulting to new standalone units. Fourth, apply climate, health, tenure, accessibility, and lifecycle criteria together. Fifth, combine weighted comparison with non-compensable safety and rights gates. The author-developed SHELTER framework offers one structure for doing so. Its value will depend on transparent local weighting, resident power, rigorous measurement, and empirical validation across diverse settings.



## Declarations

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**Consent for publication:** Not applicable.

**Availability of data and materials:** No new dataset was generated. All evidence sources are listed in the references. The conceptual figures and framework were created for this manuscript.

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